

Solution Requirements

Date	02 July 2026
Team ID	XXXXXX
Project Name	Smart Lender - AI Powered Loan Eligibility Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Functional Requirements (FR) define specific behaviors, functions, or features of the system (what the system should do). Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (how the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	User Input	Applicant enters personal, income and loan details through a web form.	High
2	Input Validation	System validates that all required numeric and categorical fields are supplied before prediction.	High
3	Feature Encoding	Categorical fields (Gender, Married, Education, etc.) are label-encoded using saved encoders.	High
4	Feature Scaling	Numeric features are standardized using a saved StandardScaler before prediction.	High
5	Loan Prediction	Trained KNN model predicts loan status as Approved or Rejected.	High
6	Confidence & Risk Scoring	System reports a prediction confidence percentage and a Low/Medium/High risk level.	Medium
7	Result Reporting	Result page displays prediction, confidence, algorithm used, model accuracy, and applicant summary.	High
8	Deployment Support	Application must run via Flask locally and be deployable using Unicorn on a cloud host.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction should be generated quickly after form submission.	< 2 seconds
2	Scalability	Application should support increasing numbers of concurrent applicants.	1000+ users
3	Security & Data Privacy	Applicant financial data must be validated and not persisted beyond the session.	No data stored
4	Reliability & Availability	System should remain available for prediction requests.	99% uptime

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
5	Usability & Accessibility	Simple, responsive interface for both desktop and mobile users.	Mobile-friendly UI
6	Maintainability	Model and preprocessing artifacts can be retrained/updated without code changes.	Modular design